

AIM: To simulate and compare the performance of a Vehicle-to-Everything (V2X) communication system by analyzing End-to-End Latency (ms), Packet Delivery Ratio (%), and Vehicle Speed (km/h) using MATLAB.

Objective:

1. To evaluate the End-to-End Latency (ms) of a V2X communication system under different vehicle operating conditions.
2. To analyze the Packet Delivery Ratio (%) and determine the reliability of V2X communication during data transmission.
3. To study the impact of Vehicle Speed (km/h) on the communication performance and stability of the V2X network.

Procedure

1. Define the V2X communication parameters such as vehicle speed, transmission range, packet size, and simulation time.
2. Generate different vehicle movement scenarios by varying the vehicle speed.
3. Simulate wireless communication between vehicles using a V2X communication model.
4. Calculate the End-to-End Latency (ms) for data transmission between communicating vehicles.
5. Compute the Packet Delivery Ratio (PDR %) by comparing the number of successfully received packets with transmitted packets.
6. Analyze the effect of increasing Vehicle Speed (km/h) on communication latency and packet delivery performance.
7. Plot MATLAB graphs for End-to-End Latency, Packet Delivery Ratio, and Vehicle Speed.
8. Compare the obtained results to evaluate the efficiency and reliability of the V2X communication system under different traffic conditions.

Objective 1 Matlab Code and Output:

```
clc;
clear;
close all;
% Vehicle Speed (km/h)
speed = 20:20:120;
% Simulated End-to-End Latency (ms)
latency = [18 16 14 12 10 9];
disp('Vehicle Speed (km/h) End-to-End Latency (ms)')
disp([speed' latency'])
```

```
figure
% ----- Graph 1 -----
subplot(2,1,1)
plot(speed,latency,'-o','LineWidth',2)
title('Vehicle Speed vs End-to-End Latency')
xlabel('Vehicle Speed (km/h)')
ylabel('Latency (ms)')
grid on
% ----- Graph 2 -----
subplot(2,1,2)
plot(1:length(latency),latency,'-s','LineWidth',2)
title('Latency Variation Across Vehicles')
xlabel('Vehicle Index')
ylabel('End-to-End Latency (ms)')
grid on
```

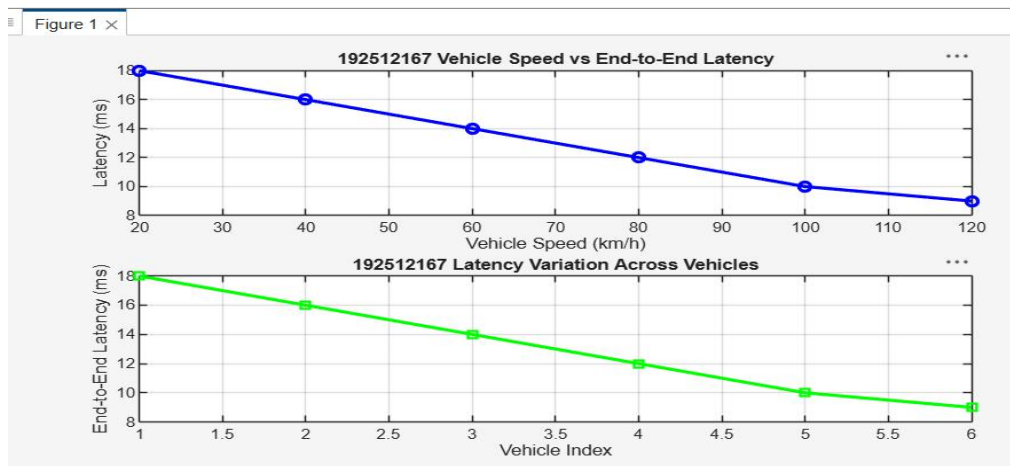
```

/MATLAB Drive/exp121.m
1  clc;
2  clear;
3  close all;
4  % Vehicle Speed (km/h)
5  speed = 20:20:120;
6  % Simulated End-to-End Latency (ms)
7  latency = [18 16 14 12 10 9];
8  disp('Vehicle Speed (km/h) End-to-End Latency (ms)')
9  disp([speed' latency'])
10
11  figure
12  % ----- Graph 1 -----
13  subplot(2,1,1)
14  plot(speed,latency,'-o','LineWidth',2,'Color','b')
15  title('192512167 Vehicle Speed vs End-to-End Latency')
16  xlabel('Vehicle Speed (km/h)')
17  ylabel('Latency (ms)')
18  grid on
19  % ----- Graph 2 -----
20  subplot(2,1,2)
21  plot(1:length(latency),latency,'-s','LineWidth',2,'Color','g')
22  title('192512167 Latency Variation Across Vehicles')
23  xlabel('Vehicle Index')
24  ylabel('End-to-End Latency (ms)')

```

Command Window

60	14
80	12
100	10
120	9



The End-to-End Latency decreases as the vehicle speed increases under the simulated communication conditions. Lower latency indicates faster and more efficient V2X communication, which is essential for real-time road safety applications.

Objective 2 Matlab Code and Output:

```

clc;
clear;
close all;
% Vehicle Speed (km/h)

speed = 20:20:120;
% Simulated Packet Delivery Ratio (%)
PDR = [99 98 97 95 93 90];
disp('Vehicle Speed (km/h) Packet Delivery Ratio (%)')
disp([speed' PDR'])
figure
% ----- Graph 1 -----
subplot(2,1,1)
plot(speed,PDR,'-o','LineWidth',2)
title('Vehicle Speed vs Packet Delivery Ratio')
xlabel('Vehicle Speed (km/h)')
ylabel('Packet Delivery Ratio (%)')
grid on
% ----- Graph 2 -----

```

```

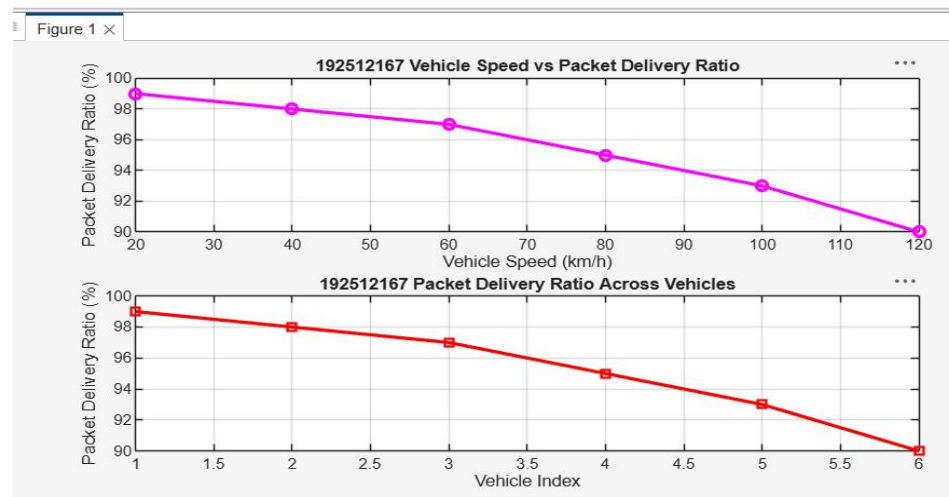
subplot(2,1,2)
plot(1:length(PDR),PDR,'-s','LineWidth',2)
title('Packet Delivery Ratio Across Vehicles')
xlabel('Vehicle Index')
ylabel('Packet Delivery Ratio (%)')
grid on

```

```

MATLAB Drive
exp10.m × exp101.m × exp103.m × EXP111.m × exp112.m × exp113.m × exp121.m × +
/MATLAB Drive/exp121.m
10 disp([speed' PDR' ])
11 figure
12 % ----- Graph 1 -----
13 subplot(2,1,1)
14 plot(speed,PDR,'-o','LineWidth',2,'Color','magenta')
15 title('192512167 Vehicle Speed vs Packet Delivery Ratio')
16 xlabel('Vehicle Speed (km/h)')
17 ylabel('Packet Delivery Ratio (%)')
18 grid on
19 % ----- Graph 2 -----
20 subplot(2,1,2)
21 plot(1:length(PDR),PDR,'-s','LineWidth',2,'Color','r')
22 title('192512167 Packet Delivery Ratio Across Vehicles')
23 xlabel('Vehicle Index')
24 ylabel('Packet Delivery Ratio (%)')
25 grid on

```



The Packet Delivery Ratio decreases slightly as vehicle speed increases due to the higher mobility of vehicles. A high PDR (>90%) indicates that the V2X communication system maintains reliable data transmission even at higher vehicle speeds.

Objective 3 Matlab Code and Output:

```

clc;
clear;
close all;
% Vehicle IDs
vehicle = 1:6;
% Vehicle Speed (km/h)
speed = [30 45 60 75 90 105];
% Communication Range (m)
range = [280 260 240 220 200 180];
disp('Vehicle Speed(km/h) Communication Range(m)')
disp([vehicle' speed' range'])
figure
% ----- Graph 1 -----
subplot(2,1,1)
plot(vehicle,speed,'-o','LineWidth',2)
title('Vehicle Speed of Different Vehicles')
xlabel('Vehicle Index')

```

```

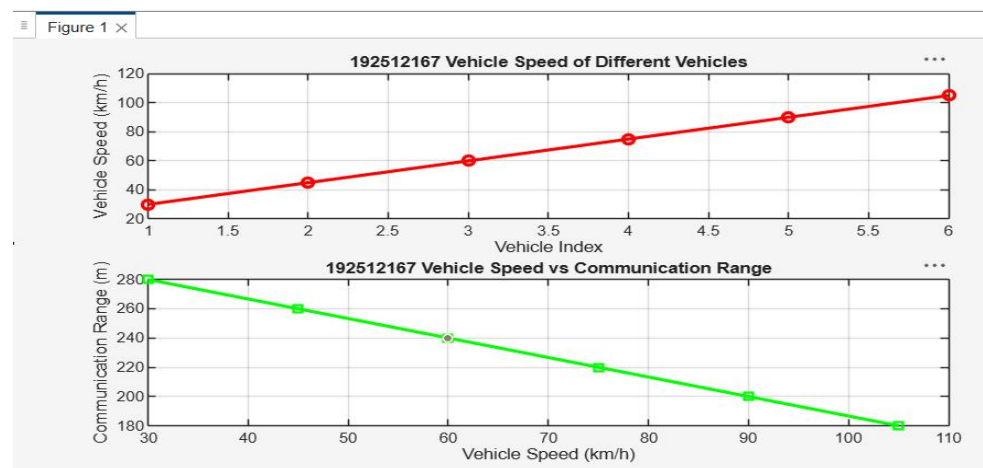
ylabel('Vehicle Speed (km/h)')
grid on
% ----- Graph 2 -----
subplot(2,1,2)
plot(speed,range,'-s','LineWidth',2)
title('Vehicle Speed vs Communication Range')
xlabel('Vehicle Speed (km/h)')
ylabel('Communication Range (m)')
grid on

```

```

/MATLAB Drive/exp121.m
1  clc;
2  clear;
3  close all;
4  % Vehicle IDs
5  vehicle = 1:6;
6  % Vehicle Speed (km/h)
7  speed = [30 45 60 75 90 105];
8  % Communication Range (m)
9  range = [280 260 240 220 200 180];
10 disp(' Vehicle Speed(km/h) Communication Range(m)')
11 disp([vehicle' speed' range'])
12 figure
13 % ----- Graph 1 -----
14 subplot(2,1,1)
15 plot(vehicle,speed,'-o','LineWidth',2,'Color','r')
16 title('192512167 Vehicle Speed of Different Vehicles')
17 xlabel('Vehicle Index')
18 ylabel('Vehicle Speed (km/h)')
19 grid on
20 % ----- Graph 2 -----
21 subplot(2,1,2)
22 plot(speed,range,'-s','LineWidth',2,'Color','g')
23 title('192512167 Vehicle Speed vs Communication Range')
24

```



The first graph shows the variation in speed among different vehicles in the V2X communication network.

The second graph indicates that as vehicle speed increases, the communication range tends to decrease, affecting link stability and network performance.

Result:

1. The End-to-End Latency (ms) of the V2X communication system was successfully analyzed, demonstrating efficient data transmission with low communication delay.
2. The Packet Delivery Ratio (PDR %) was evaluated and showed high packet delivery reliability under different vehicle operating conditions.
3. The impact of Vehicle Speed (km/h) on V2X communication was successfully studied, indicating that higher vehicle speeds can influence communication range, latency, and overall network performance.